

Project Proposal: Design of a Vision-Language Model for 3D Medical Image Analysis

Project title	Design of a Vision-Language Model for 3D Medical Image Analysis
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Course	Deep Learning — Mini Project (PhD Coursework)
Base papers	Med3DVLM: An Efficient Vision-Language Model for 3D Medical Image Analysis [5]; Diagnostic Captioning by Cooperative Task Interactions and Sample-Graph Consistency [6]
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This project designs a small vision-language model that recognises organs in 3D CT scans by matching each scan with a text description of the organ.

1. Introduction

Computed tomography (CT) produces 3D images of the body, and recognising the organs in these images is a basic step in medical image analysis. Deep learning models, especially 3D convolutional neural networks (CNNs), can learn to do this automatically.

A standard CNN learns only from class numbers (0, 1, 2, ...) and does not know what the classes mean. A vision-language model (VLM) learns from images and text together, so it also learns the meaning of words such as “liver” or “left kidney”. Recent 3D medical VLMs such as Med3DVLM [5] show strong results, but they need very large datasets and many GPUs.

This project builds a small and simple 3D VLM that can be trained on a single GPU, using a public CT dataset of 11 organs.

2. Objective

To design and train a simple vision-language model that classifies 3D CT scans into 11 organ classes by matching each scan with a text description of the organ, and to compare it with a standard 3D CNN classifier.

3. Literature survey

Paper	Method	Relevance to this project
MedMNIST v2 — Yang et al., 2023 [1]	A benchmark of small, standardised 2D and 3D medical images with CNN baselines such as ResNet	Provides the dataset and the CNN baseline
CLIP — Radford et al., 2021 [2]	Trains an image encoder and a text encoder together; an image is classified by finding the most similar text description	Core idea of the proposed model
SigLIP — Zhai et al., 2023 [3]	Replaces CLIP's softmax loss with a pairwise sigmoid loss that works well with small batches	Training loss of the proposed model
M3D — Bai et al., 2024 [4]	A large 3D CT vision-language model (M3D-LaMed) trained on 120,000 image-text pairs	Shows that VLMs work on 3D CT, but needs heavy computation
Med3DVLM — Xin et al., 2026 [5] (base paper)	An efficient 3D VLM using a lightweight 3D encoder (DCFormer) and SigLIP image-text alignment; improves image-text retrieval Recall@1 from 19.10% to 61.00% over M3D-LaMed	Image-text alignment approach followed in this project
Diagnostic Captioning — Wang et al., 2025 [6] (base paper)	Adds an image-text matching task to medical report generation so the image features capture fine clinical differences	Shows that aligning image and text features improves medical image understanding

Research gap: existing 3D medical vision-language models need large datasets and multiple high-end GPUs. This project studies whether a small 3D VLM, trainable on a single GPU, can match or improve a standard 3D CNN on organ classification.

4. Dataset

The project uses **OrganMNIST3D**, a public 3D CT dataset from the MedMNIST v2 benchmark [1].

Item	Details
Dataset	OrganMNIST3D (MedMNIST v2) [1]
Image type	3D abdominal CT volumes, grayscale
Source	201 CT scans from the Liver Tumor Segmentation (LITS) benchmark, cropped around each organ
Volume size	$28 \times 28 \times 28$ voxels
Total no. of samples	1,742
Training / validation / test	971 / 161 / 610
No. of classes	11
Class names	Liver, right kidney, left kidney, right femur, left femur, bladder, heart, right lung, left lung, spleen, pancreas
Licence and access	CC BY 4.0; downloaded with the <code>medmnist</code> Python package

The official training, validation and test split is used, so the results can be compared with published MedMNIST baselines.

5. Methodology

The existing method is a standard 3D CNN classifier, as used in the MedMNIST v2 benchmark [1]. It is built first and serves as the baseline.

- 1. Data preparation.** Load OrganMNIST3D with the `medmnist` package and scale voxel values to the range 0–1. During training, apply random flips along the three axes to increase variety.
- 2. 3D CNN model.** Four convolution blocks, each with a $3 \times 3 \times 3$ convolution, batch normalisation, ReLU activation and max pooling, with 32, 64, 128 and 256 filters. Global average pooling and a fully connected layer then give 11 class scores. The model has about 1.17 million parameters.
- 3. Training.** Cross-entropy loss, Adam optimiser, learning rate 0.001, batch size 32, 50 epochs. The model with the best validation accuracy is kept.

4. **Testing.** Measure accuracy and AUC on the 610 test volumes.

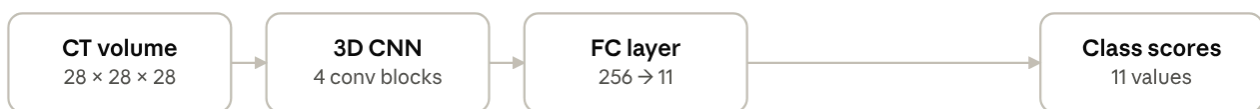
Tools: Python, PyTorch, Hugging Face Transformers and the `medmnist` package. Training runs on a single GPU; a free Google Colab GPU is sufficient for this dataset size.

6. Modification in methodology

The modification turns the 3D CNN into a simple vision-language model, following the image-text alignment idea of the base paper Med3DVLM [5]. The final fully connected layer is replaced by matching each scan against text descriptions of the 11 organs.

The proposed model predicts the organ whose text description best matches the scan

Baseline: standard 3D CNN



Proposed: simple vision-language model (new parts highlighted)

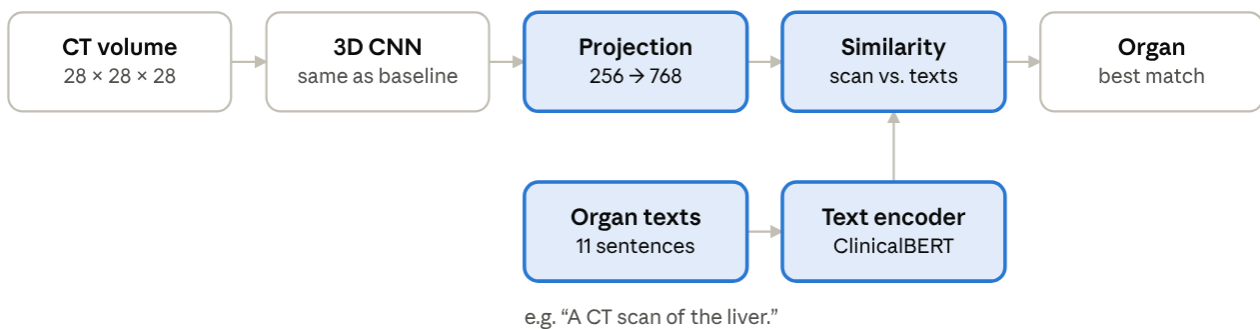


Figure 1. Baseline 3D CNN and the proposed vision-language model

1. **Organ descriptions.** Write one short sentence for each organ, for example "A CT scan of the liver." This gives 11 sentences.
2. **Text encoder.** A pretrained ClinicalBERT model [7] turns each sentence into a vector of 768 numbers. It is kept frozen, so it is not retrained. Because the 11 sentences differ by only one word, their raw vectors are almost identical, so the average of the 11 vectors is subtracted from each one (centring) to make them distinct.
3. **Image encoder.** The same 3D CNN as the baseline, followed by a projection layer that maps its 256 features to a vector of 768 numbers.
4. **Training.** The SigLIP loss [3] pulls each scan's vector towards the vector of its correct organ sentence and pushes it away from the other 10 sentences.
5. **Prediction.** The organ whose sentence vector is most similar to the scan vector is the predicted class.

The training loss and the prediction rule are given below. For scan i and organ sentence j , s is the cosine similarity of their vectors, z is +1 for the correct organ and -1 otherwise, σ is the sigmoid function, N is the number of scans in a batch, and t and b are learnable scale and bias values. The predicted organ is the sentence with the highest similarity.

$$\mathcal{L}_{\text{SigLIP}} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{11} \log \sigma(z_{ij}(t s_{ij} + b))$$

$$\hat{y}_i = \arg \max_j s_{ij}$$

Aspect	Baseline (existing method)	Proposed (modified method)
Labels used	Class numbers 0-10	A text description of each organ
Final layer	Fully connected layer with 11 outputs	Projection layer + similarity with 11 text vectors
Loss function	Cross-entropy	SigLIP (sigmoid image-text loss)
Model type	Image-only CNN	Vision-language model
Trainable parameters	About 1.17 million	About 1.36 million (text encoder frozen)

This modification lets the model learn the meaning of each organ name instead of a bare class number, applies the base paper's image-text alignment at a scale that one GPU can train, and allows new text descriptions to be added later without changing the network.

7. Evaluation and expected outcome

The baseline and the proposed model are trained on the same 971 training volumes and tested on the same 610 test volumes.

- **Accuracy:** percentage of test volumes assigned to the correct organ.
- **AUC:** area under the ROC curve, averaged over the 11 classes.
- **Confusion matrix:** shows which organs are confused with each other, for example the left and right kidney.

Expected outcome: a working 3D vision-language model for organ recognition that reaches test accuracy comparable to the standard 3D CNN, together with a comparison of the two approaches.

References

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